

Grade 6 Accelerated
Unit 11
Lesson 6 Homework

Name _____
Date _____
Period _____

1. Sophia purchased a new car for \$26,714.00. She read that the value of new cars change by $\frac{-4}{19}$ in the first year. What is the value of Sophia's car after the first year?

2. Sophia learns that the value of her car will continue to change by $\frac{-3}{17}$ of its value each year after the first year. What is the value of Sophia's car at the end of year 2?

3. An Orca whale is swimming $-250\frac{3}{4}$ ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 183.74 feet per hour. How long, in hours, will it take the Orca to reach his pod?

4. A candle store sells the 3-wick candles for $3\frac{1}{3}$ times what they paid for them from the warehouse. The candles are purchased from the warehouse at \$6.75. How much profit does the candle store make off each candle?

5. After 6 months, the price of the candles is reduced by $\frac{1}{5}$ of the original store price. What is the price of the candles after the 6 month reduction?

6. Priscilla is planning to hike a new trail this weekend. She sets up camp Friday evening at the start of the trail which begins at an elevation of -184.56 ft and ends at an elevation of 759.22 ft. She plans to begin her hike in the early morning and wants to travel $\frac{5}{8}$ of the change in elevation of her trip on Saturday, then set up camp until the next morning. What is Priscilla's starting elevation on Sunday?
7. Kate is running a race on Monday, which has a total distance of 26.13 miles. Kate's family wants to show their support by holding up encouraging signs along the path. Her mother decides to stand at the spot that is $\frac{4}{13}$ of the total race while her father chooses to stand at the spot that is $\frac{10}{13}$ of the total race. How many miles apart are Kate's parents standing from each other?
8. Kate runs at a steady rate of 12.3 minutes per mile. Using the information given in Question 7, how many minutes after the race starts should her mother expect to see Kate run by?
9. Joshua used his store credit card to purchase a new basketball. He has a coupon that when he uses his credit card, the item will be $\frac{-1}{3}$ of the total. The basketball was priced at $\$21.39$. A month goes by and Joshua forgot to pay his credit card bill, so they have added a fee of 0.29 times the amount he owed. What is the total amount Joshua has to pay his credit card company?
10. Sandy is moving to Jacksonville, Florida from New York, New York. The total distance of her move is 945.61 miles. She rented a moving truck that charges $\$19.95$ a day and $\$0.35$ per mile over 500 miles. If she returns the truck with a full tank of gas, the company will discount her bill by $\frac{-2}{31}$. What is the total amount Sandy will pay for the moving truck if she fills the tank with gas before returning it?